NEET TEST HUB

NTH-PREMIUM TEST SERIES 09

Date – 30 November 2025	Duration – 3 Hours	Marks – 720

Physics :- Units & Measurements, Motion in a Straight Line, Motion in a Plane, Laws of Motion, Work, Energy & Power , System of Particles & Rotational Motion, Gravitation

Chemistry :- Some Basic Concepts Of Chemistry, Structure of Atom, Classification of Elements & Periodicity, Chemical Bonding, Thermodynamic

Biology :- The Living World, Biological Classification, Plant Kingdom, Animal Kingdom, Morphology of Flowering Plants, Anatomy of Flowering Plant, Structural Organisation in Animals, Cell The Unit Of Life, Biomolecules, Cell Cycle & Cell Division

Instruction

1. The test is of 3 hours duration.

2. The test booklet consists of 200 questions. The maximum mark is 720.

3. There are four Sections in the Question Paper, Sections I, II, III, and IV consisting of Section I (Physics), Section II (Chemistry), Section III (Botany) and Section IV (Zoology) have 50 Questions in each Subject and each subject is divided into two Sections,

4. Section A consists of 35 questions (all questions compulsory) and Section B consists of 15 Questions (Any 10 questions are compulsory).

5. There is only one correct response for each question.

6. Each correct answer will give 4 marks while 1 Mark will be deducted for a wrong MCQ response.

7. No student is allowed to carry any textual material, printed, or written, bits of paper, pager, mobile phone, any electronic device, etc. Inside the examination room/hall.

8. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the Candidates are allowed to take away this Test Booklet with them

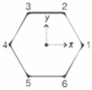
Neet Test Hub Premium Test 09

Physics

1. If masses of two point objects are tripled and distance between them is doubled, then gravitational force of attraction between them will .... $\%$

A) Increase by 225%
B) Decrease by 56%
C) Increase by 125%
D) Decrease by 144%

2. Six objects are placed at the vertices of a regular hexagon. The geometric centre of the hexagon is at the origin with objects 1 and 4 on the X-axis (see figure). The mass of the k th object is $m_k = k^i M |\cos\theta_k|$, where i is an integer, M is a constant with dimension of mass and θ_k is the angular position of the k th vertex measured from the positive X-axis in the counter-clockwise sence If the net gravitational force on a body at the centroid vanishes, the value of i is



A) 0 B) 1 C) 2 D) 3

3. If the mass of the planet is 10% less than that of the earth and radius is 20% greater than that of earth, the acceleration due to gravity on the planet will be

A) $\frac{5}{8}$ times that on the surface of earth
B) $\frac{3}{4}$ times that on the surface of earth
C) $\frac{1}{2}$ times that on the surface of earth
D) $\frac{9}{10}$times that on the surface of earth

4. Let ω be the angular velocity of the earth's rotation about its axis. Assume that the acceleration due to gravity on the earth's surface has the same value at the equator and the poles. An object weighed at the equator gives the same reading as a reading taken at a depth d below earth's surface at a pole $(d << R)$ The value of d is

A) $\frac{\omega^2 R^2}{g}$
B) $\frac{\omega^2 R^2}{2g}$
C) $\frac{2\omega^2 R^2}{g}$
D) $\frac{\sqrt{Rg}}{g}$

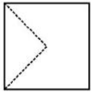
5. The gravitational force between two stones of mass $1\,kg$ each separated by a distance of $1\,metre$ in vacuum is

A) $Zero$
B) $6.675 \times 10^{-5}\,newton$
C) $6.675 \times 10^{-11}\,newton$
D) $6.675 \times 10^{-8}\,newton$

6. Two particle of masses $1\ kg$ and $3\ kg$ have position vector $2\hat{i} + 3\hat{j} + 4\hat{k}$ and $-2\hat{i} + 3\hat{j} - 4\hat{k}$ respectively. The centre of mass has a position vector

A) $-\hat{i} - 3\hat{j} - 2\hat{k}$
B) $\hat{i} + 3\hat{j} - 2\hat{k}$
C) $-\hat{i} + 3\hat{j} + 2\hat{k}$
D) $-\hat{i} + 3\hat{j} - 2\hat{k}$

7. From the homogeneous square plate we cut a triangle (Figure). Side of the square is a and. the apex of the triangle is at the center of the square. Distance fiom the center of the square to the center of mass of the remainder of the plate is



A) $a/5$
B) $a/3$
C) $a/6$
D) $a/9$

8. Consider the following statements
Assertion (A) : A cyclist always bends inwards while negotiating a curve
Reason (R) : By bending he lowers his centre of gravity Of these statements,

A) both A and R are true and R is the correct explanation of A
B) both A and R are true but R is not the correct explanation of A
C) A is true but R is false
D) A is false but R is true

9. Two masses m_1 and m_2 ($m_1 > m_2$) are connected by massless flexible and inextensible string passed over massless and frictionless pulley. The acceleration of center of mass is

A) $\left(\frac{m_1 - m_2}{m_1 + m_2}\right)^2 g$
B) $\frac{m_1 - m_2}{m_1 + m_2} g$
C) $\frac{m_1 + m_2}{m_1 - m_2} g$
D) Zero

10. Where will be the centre of mass on combining two masses m and M $(M > m)$

A) Towards m
B) Towards M
C) Between m and M
D) Anywhere

11. Identify the correct statements from the following:
(A) Work done by a man in lifting a bucket out of a well by means of a rope tied to the bucket is negative.
(B) Work done by gravitational force in lifting a bucket out of a well by a rope tied to the bucket is negative.
(C) Work done by friction on a body sliding down an inclined plane is positive.
(D) Work done by an applied force on a body moving on a rough horizontal plane with uniform velocity in zero.
(E) Work done by the air resistance on an oscillating pendulum in negative.
Choose the correct answer from the options given below:

A) B and E only
B) A and C only
C) B, D and E only
D) B and D only

12. A particle of mass m is projected at an angle α to the horizontal with initial velocity u m/s then work done by gravity during the time it reaches its highest point

A) $u^2\, sin^2\,\alpha$
B) $\frac{mu^2\cos^2\alpha}{2}$
C) $\frac{-mu^2\sin^2\alpha}{2}$
D) $\frac{mu^2\,g\sin^2\alpha}{2}$

13. A horizontal force of $5\ N$ is required to maintain a velocity of $2\ m/s$ for a block of $10\ kg$ mass sliding over a rough surface. The work done by this force in one minute is....J

A) 600
B) 60
C) 6
D) 6000

14. A $60\,kg$ weight is dragged on a horizontal surface by a rope upto 2 metres. If coefficient of friction is $\mu = 0.5$, the angle of rope with the surface is 60° and $g = 9.8\ m/\sec^2$, then work done is .... $joules$

A) 294
B) 315
C) 588
D) 197

15. A force $\vec{F} = 3\hat{i} + c\hat{j} + 2\hat{k}$ acting on a particle causes a displacement $\vec{S} = -4\hat{i} + 2\hat{j} - 3\hat{k}$ in its own direction. If the work done is $6J$, then the value of c will be

A) 12
B) 6
C) 1
D) 0

16. A uniform rope lies on a horizontal table so that a part of it hangs over the edge. The rope begins to slide down when the length of the hanging part is 25% of the entire length. The coefficient of friction between the rope and the table is

A) 0.25
B) 0.75
C) 0.33
D) 0.67

17. A block of mass M is being pulled along rough horizontal surface. The coefficient of friction between the block and the surface is μ. If another block of mass $M/2$ is placed on the block and it is again pulled on the surface, the coefficient of friction between the block and the surface will be

A) μ
B) $3\mu/2$
C) 2μ
D) $5\mu/2$

18. A block of mass m is placed on a surface having vertical cross section given by $y = x^2/4$. If coefficient of friction is 0.5 , the maximum height above the ground at which block can be placed without slipping is:

A) $1/4\ \mathrm{m}$
B) $1/2\ \mathrm{m}$
C) $1/6\ \mathrm{m}$
D) $1/3\ \mathrm{m}$

19. A block of mass m rests on a rough inclined plane. The coefficient of friction between the surface and the block is μ. At what angle of inclination θ of the plane to the horizontal will the block just start to slide down the plane?

A) $\theta = \tan^{-1}\mu$
B) $\theta = \cos^{-1}\mu$
C) $\theta = \sin^{-1}\mu$
D) $\theta = \sec^{-1}\mu$

20. A car travels north with a uniform velocity. It goes over a piece of mud which sticks to the tyre. The particles of the mud, as it leaves the ground are thrown

A) Vertically upwards
B) Vertically inwards
C) Towards north
D) Towards south

21. An athlete does not come to rest immediately after crossing the winning line due to the

A) Inertia of rest
B) Inertia of motion
C) Inertia of direction
D) None of these

22. $Assertion$: Use of ball bearings between two moving parts of a machine is a common practice.
$Reason$: Ball bearings reduce vibrations and provide good stability.

A) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.
B) If both Assertion and Reason are correct but Reason is not a correct explanation of the

Assertion.

C) If the Assertion is correct but Reason is incorrect.

D) If both the Assertion and Reason are incorrect.

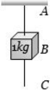
23. Newton's first law of motion describes the following

A) Energy
B) Work
C) Inertia
D) Moment of inertia

24. When a train stops suddenly, passengers in the running train feel an instant jerk in the forward direction because

A) The back of seat suddenly pushes the passengers forward
B) Inertia of rest stops the train and takes the body forward
C) Upper part of the body continues to be in the state of motion whereas the lower part of the body in contact with seat remains at rest
D) Nothing can be said due to insufficient data

25. A mass of $1kg$ is suspended by a string A. Another string C is connected to its lower end (see figure)., if the string C is stretched slowly, then



A) The portion AB of the string will break
B) The portion BC of the string will break
C) None of the strings will break
D) None of the above

26. Velocity of a particle moving in a curvilinear path in a horizontal $X\ Y$ plane varies with time as $\vec{v} = (2t\hat{i} + t^2\hat{j})\ \ m/s$. Here, t is in second. At $t = 1\ s$

A) acceleration of particle is $8\ m/s^2$
B) tangential acceleration of particle is $\frac{4}{\sqrt{5}}\ m/s^2$
C) radial acceleration of particle is $\frac{6}{\sqrt{5}}\ m/s^2$
D) radius of curvature to the path is $\frac{5\sqrt{5}}{2}\ m$

27. A particle moves in the xy plane with a constant acceleration $'g'$ in the negative y-direction. Its equation of motion is $y = ax - bx^2$, where a and b are constants. Which of the following are correct?

A) The x-component of its velocity is constant.
B) At the origin, the$y$-component of its velocity is a$\sqrt{\frac{g}{2b}}$.
C) At the origin, its velocity makes an angle $tan^{-1}(a)$ with the x-axis.
D) All of the above

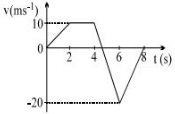
28. A particle moves in a circle of radius $25\,cm$ at two revolutions per sec. The acceleration of the particle in m/s^2 is

A) π^2
B) $8\pi^2$
C) $4\pi^2$
D) $2\pi^2$

29. A particle is projected vertically upwards from O with velocity v and a second particle is projected at the same instant from P (at a height h above O) with velocity v at an angle of projection θ . The time when the distance between them is minimum is

A) $\frac{h}{2v\sin\theta}$
B) $\frac{h}{2v\cos\theta}$
C) h/v
D) $h/2v$

30. The figure shows a velocity-time graph of a particle moving along a straight line The total distance travelled by the particle is .... m



A) 66.6
B) 51.6
C) 0
D) 36.6

31. Let θ be the angle between vectors $\vec{A}$ and $\vec{B}$. Which of the following figures correctly represents the angle θ ?

A) 
B) 
C) 
D) 

32. If $|\vec{v}_1 + \vec{v}_2| = |\vec{v}_1 - \vec{v}_2|$ and $\vec{v}_1$ and $\vec{v}_2$ are finite, then

A) $\vec{v}_1$ is parallel to $\vec{v}_2$
B) $\vec{v}_1 = \vec{v}_2$
C) $|\vec{v}_1| = |\vec{v}_2|$
D) $\vec{v}_1$ and $\vec{v}_2$ are mutually perpendicular

33. The position vectors of points A, B, C and D are $\vec{A} = 3\hat{i} + 4\hat{j} + 5\hat{k},\ \vec{B} = 4\hat{i} + 5\hat{j} + 6\hat{k},\ \vec{C} = 7\hat{i} + 9\hat{j} + 3\hat{k}$ and $\vec{D} = 4\hat{i} + 6\hat{j}$ then the displacement vectors $\overrightarrow{AB}$ and $\overrightarrow{CD}$ are

A) Perpendicular
B) Parallel
C) Antiparallel
D) Inclined at an angle of 60^o

34. $0.4\hat{i} + 0.8\hat{j} + c\hat{k}$ represents a unit vector when c is

A) -0.2
B) $\sqrt{0.2}$
C) $\sqrt{0.8}$
D) 0

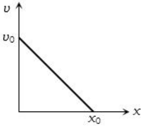
35. Two adjacent sides of a parallelogram are represented by the two vectors $\hat{i}+2\hat{j}+3\hat{k}$ and $3\hat{i}-2\hat{j}+\hat{k}$. What is the area of parallelogram

A) 8 **B)** $8\sqrt{3}$ **C)** $3\sqrt{8}$ **D)** 192

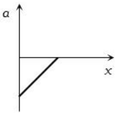
36. A particle is moving along a circle such that it completes one revolution in 40 seconds. In 2 minutes 20 seconds, the ratio $\frac{|\text{displacement}|}{\text{distance}}$ is ....

A) 0 **B)** $\frac{1}{7}$ **C)** $\frac{2}{7}$ **D)** $\frac{1}{11}$

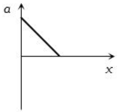
37. The given graph shows the variation of velocity with displacement. Which one of the graph given below correctly represents the variation of acceleration with displacement



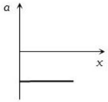
A)



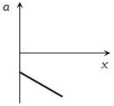
B)



C)



D)



38. A ball is thrown vertically upwards from the top of a tower with a velocity u. This ball reaches the ground level with a velocity $4\,u$. The height of the tower is

A) $\frac{3u^2}{g}$ **B)** $\frac{4u^2}{g}$
C) $\frac{6u^2}{g}$ **D)** $\frac{15u^2}{2g}$

39. Speed of two identical cars are u and $4u$ at a specific instant. The ratio of the respective distances in which the two cars are stopped from that instant is

A) $1:1$ **B)** $1:4$ **C)** $1:8$ **D)** $1:16$

40. Which of the following is a one dimensional motion

A) Landing of an aircraft
B) Earth revolving a round the sun
C) Motion of wheels of a moving trains
D) Train running on a straight track

41. A quantity f is given by $f=\sqrt{\frac{hc^5}{G}}$ where c is speed of light, G universal gravitational constant and h is the Planck's constant. Dimension of f is that of

A) Momentum **B)** Area
C) Energy **D)** Volume

42. The relative error in the determination of the surface area of a sphere is α. Then the relative error in the determination of its volume is

A) $\frac{2}{3}\alpha$ **B)** $\frac{5}{2}\alpha$
C) $\frac{3}{2}\alpha$ **D)** α

43. Which of the following is the unit of specific heat

A) $J/kg\,^{\circ}C$ **B)** $J\,kg\,^{\circ}C^{-1}$
C) $kg\,^{\circ}C/J$ **D)** $J/kg\,^{\circ}C^{-2}$

44. If V denotes the potential difference across the plates of a capacitor of capacitance C, the dimensions of $CV^2$are

A) Not expressible in MLT
B) MLT^{-2}
C) M^2LT^{-1}
D) ML^2T^{-2}

45. Which is not a unit of electric field

A) NC^{-1} **B)** Vm^{-1}
C) JC^{-1} **D)** $JC^{-1}m^{-1}$

Chemistry

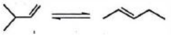
46. Calculate the enthalpy change when $50\,ml$ of $0.01\ M\ Ca(OH)_2$ reacts with $25\ ml$ of $0.01\ M\ HCl$. Given that ΔH^o for neutralisation of a strong acid and a strong base is $140\,k\,cal\,mol^{-1}$ ....$kcal$

A) 14 **B)** 35 **C)** 10 **D)** 7.5

47. An ideal gas undergoes isothermal expansion at constant pressure. During the process

A) enthalpy increases but entropy decreases
B) enthalpy remains constant but entropy increases
C) enthalpy decreases but entropy increases.
D) Both enthalpy and entropy remain constant.

48. The heat of hydrogenation for 3 -methylbutene and 2 -pentene are -30 $kcal/mol$ and -28 $kcal/mol$ respectively. The heats of combustion of 2 -methylbutane and pentane are - 784 $kcal/mol$ and -782 $kcal/mol$ respectively. All the values are given under standard conditions. Taking into account that combustion of both alkanes give the same products, what is ΔH (in $kcal/mol$) for the following reaction under same conditions ?



A) 0 B) -4 C) -2 D) 2

49. The equilibrium constant of a reaction at $298\,K$ is 5×10^{-3} and at $1000\ K$ is 2×10^{-5}. What is the sign of ΔH for the reaction

A) $\Delta H = 0$
B) ΔH is negative
C) ΔH is positive
D) None of these

50. The standard enthalpy of the decomposition of N_2O_4 to NO_2 is $58.04\,kJ$ and standard entropy of this reaction is $176.7\ J/K$. The standard free energy change for this reaction at $25\,^oC$ is .....kJ

A) 539 B) -539 C) -5.39 D) 5.39

51. The compound with negative heat of formation are known as

A) Endothermic compound
B) Exothermic compound
C) Heat of formation compound
D) None of the above

52. If a chemical reaction is accompanied by the evolution of heat, it is

A) Catalytic B) Photochemical
C) Endothermic D) Exothermic

53. Heat of combustion of a substance

A) Is always positive
B) Is always negative
C) Is equal to heat of formation
D) Nothing can be said without reaction

54. For the isothermal expansion of an ideal gas

A) E and H increases
B) E increases but H decreases
C) H increases but E decreases
D) E and H are unaltered

55. Select equation having exothermic step

A) $S^-_{(g)} \rightarrow S^{-2}_{(g)}$
B) $Na^+_{(g)} + Cl^-_{(g)} \rightarrow NaCl_{(s)}$
C) $N_{(g)} \rightarrow N^-_{(g)}$
D) $Al^{+2}_{(g)} \rightarrow Al^{+3}_{(g)}$

56. Match list I with list II and select the correct answer

list I (species)	list II ($O-N-O$ angle)
$(A)\ NO_2^+$	$(1)\ 180^o$
$(B)\ NO_2$	$(2)\ 132^o$
$(C)\ NO_2^-$	$(3)\ 120^o$
$(D)\ NO_3^-$	$(4)\ 115^o$
	$(5)\ 109^o$

A) $A-5,\ B-4,\ C-3,\ D-2$
B) $A-5,\ B-2,\ C-4,\ D-3$
C) $A-1,\ B-2,\ C-4,\ D-3$
D) $A-1,\ B-4,\ C-3,\ D-2$

57. The compound with the highest degree of covalency is

A) $NaCl$ B) $MgCl_2$ C) $AgCl$ D) $CsCl$

58. The ionic bonds X^+Y^- are formed when
(I) electron affinity of Y is high (II) ionization energy of X is low
(III) lattice energy of XY is high (IV) lattice energy of XY is low
Choose the correct code

A) I and II B) I and III
C) I, II and III D) All

59. The hydration energy of Mg^{2+} ions is lesser than that of

A) Al^{3+} B) Ba^{2+}
C) Na^+ D) None of these

60. Consider the molecules CH_4, NH_3 and H_2O. Which of the given statements is false?

A) The $H-O-H$ bond angle in H_2O is smaller than the $H-N-H$ bond angle in NH_3.
B) The $H-C-H$ bond angle in CH_4 is larger than the $H-N-H$ bond angle in NH_3.
C) The $H-C-H$ bond angle in CH_4, the $H-N-H$ bond angle in NH_3, and the $H-O-H$ bond angle in H_2O are all greater than 90^o.
D) The $H-O-H$ bond angle in H_2O is larger than the $H-C-H$ bond angle in CH_4.

61. In which of the following pairs molecules have bond order three and are isoelectronics

A) CN^-, CO B) $NO^+,\ CO^+$
C) CN^-, O_2^+ D) CO, O_2^+

62. The smallest bond angle is found in

A) IF_7 B) CH_4
C) BeF_2 D) BF_3

63. When $NaCl$ is dissolved in water the sodium ion becomes

A) Oxidized B) Reduced
C) Hydrolysed D) Hydrated

64. The law of triads is applicable to a group of

A) Cl, Br, I B) C, N, O
C) Na, K, Rb D) H, O, N

65. The electron affinity of Be is similar to :-

A) He B) B C) Li D) Na

66. Which will show maximum non-metallic character

A) B B) Be C) Mg D) Al

67. The order of the magnitude of first ionisation potentials of Be, B, N and O is

A) $N > O > Be > B$
B) $N > Be > O > B$
C) $Be > B > N > O$
D) $B > Be > O > N$

68. Which of the following ions has the smallest radius

A) Be^{2+} B) Li^{+} C) O^{2-} D) F^{-}

69. The chemistry of lithium is very similar to that of magnesium even though they are placed in different groups

A) Both are found together in nature
B) Both have nearly the same size
C) Both have similar electronic configuration
D) The ratio of their charge to size is nearly the same

70. As one moves along a given row in the periodic table, ionization energy

A) Remains same
B) Increases from left to right
C) First increases, then decreases
D) Decreases from left to right

71. Who is called the father of chemistry

A) Faraday B) Priestley
C) Rutherford D) Lavosier

72. The most importasnt active step in the development of periodic table was taken by

A) Mendeleef B) Dalton
C) Avogadro D) Cavendish

73. If $n = 2$ for He^{+} ion than ....$\overset{o}{A}$ out the wave length

A) 3.33 B) 6.42 C) 1.47 D) 2.37

74. The energy of an electron in excited hydrogen atom is $-3.4\ eV$. Then, according to Bohr's theory the angular momentum of this electron in $Jsec$ is

A) 2.1×10^{-34}
B) 4.2×10^{-34}
C) 6.6×10^{-34}
D) 3.2×10^{-33}

75. The (e/m) for positive rays in comparison to cathode rays is

A) Very low B) Very high
C) Same D) None

76. Select the correct statements for quantum numbers.
(i) Magnetic quantum no. (m_l) gives information about the spatial orientation of orbitals with respect to standard set of co-ordinate axis.
(ii) Electron spin quantum no. is represented by 's' and have value' $\frac{1}{2}$ '
(iii) Principal quantum no. (n) determine the size of the orbitals and also to a large extent of the energy of the orbitals.

A) Only $(i), (iii)$ B) Only (iii)
C) Only (i) D) $(i), (ii), (iii)$

77. The total number of electrons that can be accommodated in all the orbitals having principal quantum number 2 and azimuthal quantum number 1 is

A) 2 B) 4 C) 6 D) 8

78. For a given value of quantum number l, the number of allowed values of m is given by

A) $l + 2$ B) $2l + 2$
C) $2l + 1$ D) $l + 1$

79. The ratio of specific charge of a proton and an α-particle is

A) $2 : 1$ B) $1 : 2$ C) $1 : 4$ D) $1 : 1$

80. What is the ratio of mass of an electron to the mass of a proton

A) $1 : 2$ B) $1 : 1$ C) $1 : 1837$ D) $1 : 3$

81. Heaviest particle is

A) Meson B) Neutron C) Proton D) Electron

82. The mole fraction of urea in an aqueous urea solution containing $900g$ of water is 0.05 . If the density of the solution is $1.2gcm^{-3}$, the molarity of urea solution is. . . . . (Given data : Molar masses of urea and water are $60gmol^{-1}$ and $18gmol^{-1}$, respectively)

A) 2.50 B) 2.55 C) 2.60 D) 2.98

83. The number of significant figures in 50000.020×10^{-3} is ..... .

A) 5 B) 8 C) 2 D) 10

84. A sample of a hydrate of barium chloride weighing $61\ g$ was heated until all the water of hydration is removed. The dried sample weight is $52\ g$. The formula of the hydrated salt is : (atomic mass, $Ba = 137\, amu,\ Cl = 35.5\, amu$)

A) $BaCl_2.4H_2O$
B) $BaCl_2.3H_2O$
C) $BaCl_2.\, H_2O$
D) $BaCl_2.2H_2O$

85. The mass of carbon present in $0.5\ mole$ of $K_4[Fe(CN)_6]$ is .... g

A) 1.8 B) 18 C) 3.6 D) 36

86. The molecular weight of hydrogen peroxide is 34. What is the unit of molecular weight

A) g
B) mol
C) $g\, mol^{-1}$
D) $mol\, g^{-1}$

87. Which of the following statements is correct about equivalent weight of $KMnO_4$

A) It is one third of its molecular weight in alkaline medium
B) It is one fifth of its molecular weight in alkaline medium
C) It is equal to its molecular weight in acidic medium
D) It is one third of its molecular weight in acidic medium

88. The number of moles of sodium oxide in $620\, g$ of it is .... moles

A) 1 B) 10
C) 18 D) 100

89. The unit $J\, Pa^{-1}$ is equivalent to

A) m^3 B) cm^3
C) dm^3 D) None of these

90. The significant figures in 3400 are

A) 2 B) 5 C) 6 D) 4

Biology - (Botany)

91. A cell division in which a diploid somatic cell divides into two identical daughter cells is called

A) Meiosis I B) Meiosis II
C) Mitosis D) Cytokinesis

92. Match the following with respect to meiosis:

(a) Zygotene	(i) Terminalization
(b) Pachytene	(ii) Chiasmata
(c) Diplotene	(iii) Crossing over
(d) Diakinesis	(iv) Synapsis

Select the correct option from the following:

	(a)	(b)	(c)	(d)
A)	(ii)	(iv)	(iii)	(i)
B)	(iii)	(iv)	(i)	(ii)
C)	(iv)	(iii)	(ii)	(i)
D)	(i)	(ii)	(iv)	(iii)

93. In which cells are biosyntheticaly active but they can not divided ?

A) S phase B) G_1 phase
C) G_0 phase D) G_2 phase

94. Bacterial cell divides every one minute. It takes 15 minutes a cup to be one-fourth full. How much time will it take to fill the cup?

A) 30 minutes B) 45 minutes
C) 60 minutes D) 17 minutes

95. In which phase proteins for spindle fibre formation are synthesized

A) G_1 phase B) G_2 phase
C) $S-$ phase D) Anaphase

96. Which of the following is not strictly a biomacromolecule?

A) Proteins B) Lipids
C) Polysaccharides D) Nucleic acid

97. In the plant 'ash', this is not present.

A) Phosphate B) Sulphate
C) Cellulose D) All are present

98. About 98 percent of the mass of every living organism is composed of just six elements including carbon, hydrogen, nitrogen, oxygen and

A) sulphur and magnesium
B) magnesium and sodium
C) calcium and phosphorus
D) phosphorus and sulphur.

99. The four elements that make up 99% of all elements found in a living system are

A) H, O, C, N B) C, H, O, S
C) C, H, O, P D) C, N, O, P

100. Weight of protein is

A) > 12000 B) < 6000
C) < 12000 D) $600 - 3000$

101. Which of the following statement was not explained in the cell theory given jointly by Schleiden and Schwann?

A) All living organism are composed of cells and their products
B) Cell is the structural and functional unit of living organisms
C) Formation of new cells
D) None of the above

102. Match column-I (scientists) with column-ll (discovery) and select the correct option.

Column-I	Column-II
A. Leeuwenhoek	I. First saw and described a living cell
B. Robert Brown	II. Presence of cell wall is unique to plant cells
C. Schleiden	III. Discovered the nucleus
D. Schwann	IV. All plants are composed of different kind of cells

A) $A-I; B-III; C-IV; D-II$
B) $A-I; B-III; C-II; D-IV$
C) $A-III; B-I; C-IV; D-II$
D) $A-I; B-IV; C-II; D-III$

103. The cell theory was given in year 1839 by Schleiden and Schwann. According to this theory all organisms are composed of cell and cells are the basic unit of life. How did this theory help in the field of science?

A) It helped to study the working of cells.
B) It helped in curing diseases caused by cell.
C) It helped in restating the earlier theories on cell.
D) It helped in introducing the use of microscopes to study cell.

104. Which of the following best illustrates “feedback” in development?

A) Tissue X secretes RNA which changes the development of tissue Y.
B) As tissue X develops, it secretes enzymes that inhibit the development of tissue Y.
C) As tissue X develops, it secretes something that induces tissue Y to develop.
D) As tissue X develops, it secretes something that slows down the growth of tissue Y.

105. ....examined a large number of ....and observed that all are composed of different kinds of cells which from the tissues

A) Anton Von leuwenhoek, live cell
B) Robert brown,nucleus
C) Matthis schleiden, plant
D) Theodore and animal

106. Bark refers to all tissues exterior to the

A) Cork cambium B) Pericycle
C) Vascular cambium D) Periderm

107. Meristem is characterized by

A) Isodiametric cells with cellulosic thin wall
B) Absence of intercellular space and vacuole
C) Absence of reserve food material and plastids
D) All of these

108. In sclerenchymatous cells, the lignin deposition

A) Is sometimes absent
B) Begin from primary to secondary walls
C) Begin from secondary to primary
D) Is confined to middle lamella

109. The meristem in the root is

A) Terminal
B) Sub-apical or sub-terminal
C) Intercalary
D) Absent

110. A group of cells similar in form, function and origin is called

A) Organ B) Organella
C) Tissue D) None of these

111. Which of the following is not the lateral branches of the roots?

A) Tertiary roots
B) Secondary roots
C) Primary root
D) More than one option is correct

112. Androeium is epipetalous in this family

A) Solanaceae B) Brassicaceae
C) Liliaceae D) None

113. The calyptrogen of the root apex forms

A) Rhizoids B) Root nodule
C) Root hairs D) Root cap

114. Aestivation of corolla in Pea is

A) Contorted
B) Valvate
C) Imbricate
D) Vexillary

115. Monocot plants are characterised by the presence of

A) Tap roots B) Fibrous roots
C) Annulated roots D) Stilt roots

116. Choose the correct statements about protonema

A) Juvenile stage of moss is protonema
B) It consists of slender; green, branching system of filaments
C) Develops directly from a spore
D) All of the above

117. Kingdom-Plantae includes

A) Algae, bryophytes and pteridophytes
B) Algae, bryophytes, pteridophytes, gymnosperms and angiosperms
C) Algae, fungi, peteridophytes, gymnosperms and angiosperms
D) Algae, pteridophytes, gymnosperms and angiosperms

118. Karyotaxonomy is the modern branch of classification which is based on

A) Number of chromosomes
B) Bands found on chromosomes
C) Organic evolution
D) Trinomial nomenclature

119. Which of the following form contain algae

A) Equisetum B) Selaginella
C) Marsilea D) None of these

120. Natural system of classification differs from artifical system in

A) Employing only one floral trait
B) Taking only one vegetative trait
C) Bringing out similarities and dissimilarities
D) Developing evolutionary trends

121. Genus Penicillium belong to the class

A) Basidiomycetes B) Ascomycetes
C) Phycomycetes D) Deuteromycetes

122. Which of the following is a nitrogen fixer

A) Ulothrix B) Anabaena
C) Ulva D) Hydrodictyon

123. Latest classification of biological kingdoms has been proposed by

A) Linnaeus B) Haeckel
C) Whittaker D) John Ray

124. Phenetic classification of organisms is based on

A) Dendogram based on DNA characteristics
B) Sexual characteristics
C) Observable characteristics of existing organisms
D) The ancestral lineage of existing organisms

125. The scientist who created the group Protista for both unicellular plants and animals is

A) Haeckel B) Pasteur C) Lister D) Koch

126. Delete odd one for wheat.

A) Poaceae B) Monocotyledonae
C) Angiospermae D) Dicotyledonae

127. Which of the following taxonomic category of housefly is incorrectly matched?

A) Genus - Musca B) Family - Muscidae
C) Order - Primata D) Class - Insecta

128. Potato and brinjal belong to the genus Solanum, which reflects that

A) They belong to single species
B) They are a group of related species
C) They both are morphologically and structurally similar to each other in all respects
D) They can always produce fertile hybrid

129. Which of the following is incorrect for reproduction?

A) Unicellular organisms reproduce by cell division
B) Reproduction is a characteristic of all living organisms
C) In unicellular organisms, reproduction and growth are linked together
D) Non-living objects are incapable of reproducing

130. The fungi, the filamentous algae, the protonema of mosses, all easily multiply by____.

A) budding B) fission
C) regeneration D) fragmentation

131. Which of the following are unique features of living organisms?

A) Growth and reproduction
B) Reproduction and ability to sense environment
C) Metabolism and interaction
D) All of the above1

132. Rice, cereals, monocots and plants represent

A) Different taxa at different level
B) Same taxa of different category
C) Different category of same taxa
D) Same category for different taxa

133. In a hierarchical system of plant classification, which one of the following taxonomic ranks generally ends in '$ceae$'

A) Family B) Genus
C) Order D) Class

134. A scientist having made significant contribution in the field of classification is

A) Pasteur B) Oparin C) Darwin D) Linnaeus

135. In Botanical nomenclature of plants

A) Genus is written after the species
B) Both in genus and species the first letter is a capital letter
C) Genus and species may be same name
D) Both genus and species are printed in italics

Biology - (Zoology)

136. Match List I with List II.

List I	List II
A. Mast cells	I. Ciliated epithelium
B. Inner surface of bronchiole	II. Areolar connective tissue
C. Blood	III. Cuboidal epithelium
D. Tubular parts of nephron	IV. Specialised connective tissue

Choose the correct answer from the options give below:

A) A-III, B-IV, C-II, D-I
B) A-I, B-II, C-IV, D-III
C) A-II, B-III, C-I, D-IV
D) A-II, B-I, C-IV, D-III

137. Match the following:

$Column - I$	$Column - II$
$A.\ Caecum\ in\ earthworm$	$1.\ Tubular$
$B.\ Caecum\ in\ cockroach$	$2.\ Forgut$
$C.\ Gizzard\ in\ earthworm$	$3.\ 26^{th}\ segment$
$D.\ Gizzard\ in\ cockroach$	$4.\ 8-9\ segment$

A) $A-1, B-4, C-2, D-3$
B) $A-4, B-2, C-1, D-3$
C) $A-3, B-1, C-4, D-2$
D) $A-3, B-1, C-2, D-3$

138. Which of the following characteristics is incorrect with respect to cockroach?

A) A ring of gastric caeca is present at the junction of midgut and hind gut.
B) Hypopharynx lies within the cavity enclosed by the mouth parts.
C) In females, $7^{\text{th}} - 9^{\text{th}}$ sterna together form a genital pouch.
D) 10^{th} abdominal segment in both sexes, bears a pair of anal cerci.

139. Consider the following statements regarding cockroach and mark the correct option.
(i) Head is formed by the fusion of 6 -segments.
(ii) Mouth parts are biting and chewing type.
(iii) Crop is the part of mid gut.

A) Only (ii) is correct
B) (ii) and (iii) are correct
C) (i) and (ii) are correct
D) Only (iii) is correct

140. Lack of blood supply and presence of the noncellular basement membrane are the characteristics of the

A) muscular tissue
B) fluid connective tissue
C) epithelial tissue
D) nervous tissue

141. Match the following columns.

$Column - I$	$Column - II$
$A.\ Areolar\ tissue$	$1.\ Fat\ cells$
$B.\ Adipose\ tissue$	$2.\ Osteocytes$
$C.\ Ligament$	$3.\ Loose\ connective\ tissue$
$D.\ Bone$	$4.\ Dense\ regular\ connective\ tissue$

A) $A-3, B-1, C-4, D-2$
B) $A-1, B-2, C-3, D-4$
C) $A-4, B-3, C-2, D-1$
D) $A-2, B-1, C-4, D-3$

142. A : Intercellular material is minimum between the cells of epithelial tissue.
R : Epithelial cells are not secretory in nature.

A) Assertion and Reason both are correct and also correct explanation.
B) Assertion and Reason both are correct but not explanation of assertion.
C) Assertion is correct, but Reason is incorrect.
D) Both Assertion and Reason are incorrect.

143. Mark the correct option with respect to tissue, location, and its function.

A)

Tissue	Location	Function
Reticular issue	Spleen	Secretion

B)

Tissue	Location	Function
Brush bordered cuboidal epithelium	PCT	Reabs ortion

C)

Tissue	Location	Function
Neurosensory epithelium	Taste buds and cornea	Conversi on of all types pf electrical stimuli to chemical stimuli

D)

Tissue	Location	Function
Glandular epithelium	Lining of blood vessels	To handle blood pressure

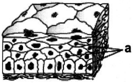
144. Non-ciliated simple columnar epithelium often contains which increase the surface area for secretion and absorption.

A) flagella
B) collagen fibres
C) microvilli
D) all of these

145. Body cavity of cockroach is also called

A) Blood cavity
B) Genital chamber
C) Heart chamber
D) Cloaca

146. Identify given figure and indicated part of 'a'.



A) Ciliated epithelium - nucleus
B) Compound epithelium -multilayered cells
C) Compound epithelium - multicellular gland
D) Ciliated epithelium -multilayered cells.

147. What external changes are visible after the last moult of a cockroach nymph?

A) Both forewings and hindwings develop
B) Labium develops
C) Mandibles become harder
D) Anal cerci develop

148. Muscular tissue is differentiated into

A) Unstriped, striped
B) Striped, cardiac
C) Cardiac muscle, unstriped
D) Unstriped, striated and cardiac

149. Identify the tissue shown in the diagram and match with its characteristics and its location.

A) Smooth muscles, show branching, found in the wall of the heart
B) Cardiac muscles, unbranched muscles, found in the walls of the heart
C) Striated muscles, tapering at bothends, attached with the bones of the ribs
D) Skeletal muscles show striations and are closely attached with the bones of the limbs

150. In frog, respiration occurs by

A) Lungs
B) Trachae
C) Gills only
D) Both (a) and (b)

151. Which of the following is not a median part in the brain cavity of frog

A) Iter
B) Diocoel
C) Metacoel
D) Optocoel

152. Ninth cranial nerve of frog is

A) Trochlear
B) Trigeminal
C) Oculomotor
D) Glossopharyngeal

153. Inner surface of fallopian tubes, bronchi and bronchioles are lined by

A) Squamous epithelium
B) Ciliated epithelium
C) Columnar epithelium
D) Cubical epithelium

154. Frog is

A) Arboreal
B) Terrestrial
C) Fully aquatic
D) Both aquatic and terrestrial

155. The intestine and stomach in mammals are lined by

A) Cuboidal epithelium
B) Columnar epithelium
C) Squamous epithelium
D) Stratified epithelium

156. Growth of young cartilage takes place by

A) Division of young chondrocytes
B) Formation of more intercellular substance
C) Deposition of new layer of cartilage at its surface
D) All of the above

157. External ear $(pinna)$ contains a hard, flexible structure composed of

A) Bone
B) Cartilage
C) Tendon
D) Ligament

158. In closed circulatory system

A) The cells and tissues are directly bathed in the blood pumped out by heart
B) Arteries and veins are lacking
C) The capillaries are largest blood vessels and closed at their ends
D) Blood circulates through a series of vessels of varying diameters

159. The only incorrectly matched pair is

Phylum	Level of organisation

A) Porifera -Cellular level

B) Cnidaria -Tissue level
C) Annelida -Organ level only
D) Mollusca -Organ system level

160. Select the incorrect option with respect to features present in three phylum
Characters -Arthropoda -Mollusca - Chordata

A) Coelom -Absent -Absent - Absent
B) Segmentation -Present -Absent - Present
C) Digestive system - Complete - Complete - Complete
D) Circulatory system -Present - Present - Present

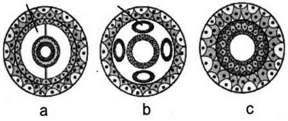
161. Three types of body cavity are

A) True coelom, pseudocoelom and haemocoel
B) Pseudocoelom, protocoelom and acoelom
C) Acoelom, deuterocoel and homocoel
D) Protocoel, deuterocoel and pseudocoelom

162. Animal are classified on the basis of which of the following features?
I.Coelomic cavity II. Level of organisation III. Notochord IV. Skeletal structure

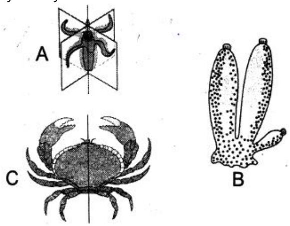
A) I and II
B) I and III
C) I, II and III
D) II and IV

163. According to given figure a, b, c choose correct answer for coelomateO



A) a - Aschelminthes, b - Platyhelminthes c - arthropods
B) a - arthropods, b - Platyhelminthes c- Aschelminthes
C) a - arthropods, b - Aschelminthes c- Platyhelminthes
D) a - Platyhelminthes, b - Aschelminthes c- arthropods

164. Which one of the following figure shows radial symmetry



A) A
B) B
C) C
D) Give all

165. Choose correct pair :
class - Character

A) Mammal - Poikilotherms
B) Reptile - Homoiothermous
C) Amphibians - Oviparous and external fertilization
D) Aves - Three chambered heart

166. Metameric segmentation is the characteristic of

A) mollusca and chordata
B) platyhelminthes and arthropoda
C) echinodermata and annelida
D) annelida and arthropoda.

167. The term for body cavity present in triploblastic animals is

A) Haemocoel
B) Pseudocoelom
C) Coelom
D) All the above

168. In which of the following jaws are found

A) Herdmania
B) Fish
C) Petromyzon
D) Amphioxus

169. The first true organ system grade body organisation is developed in

A) annelida
B) Coelenterata
C) Platyhelminthes
D) Arthropoda

170. The haemocoel of an insect is actually a

A) Modified blood vessel
B) True coelom
C) Pseudocoelom
D) Schizocoelom

171. Leech secretes which of the following anticoagulant

A) Hirudin
B) Heparin
C) Serotonin
D) Histamine

172. All worms are

A) Triploblastic B) Segmented
C) Endo-parasites D) Free-living

173. A true coleom is absent in phylum

A) Nematoda B) Annelida
C) Echinodermata D) Mollusca

174. Phylum coelenterata has remained at

A) Cellular level of organisation
B) Organ level of organisation
C) Tissue level of organisation
D) Organ system level of organisation

175. Which one is not typical to all porifers

A) Perforated body
B) Choanocytes
C) System of pores and canal
D) Presence of spongin fibres

176. Body cavity lined by mesoderm is called

A) Coelenteron B) Pseudocoel
C) Coelom D) Blastocoel

177. Body does not show any segmentation in

A) Frog B) Cockroach
C) Earthworm D) Star Fish/Hydra

178. Hydra is

A) Coral B) Worm C) Polyp D) Medusa

179. Which group is commonly called as 'sea stick'

A) Arthropoda B) Echinodermata
C) Coelenterata D) Porifera

180. Level of organisation in sponges is

A) Cellular level
B) Acellular level
C) Tissue level
D) Organ-system level